Straw Reaper



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 4 labs

Covers these lab items: Straw Reaper

What it is used for

A straw reaper processes harvested crop residue to recover straw and prepare it for collection or further use. Correct machine adjustment, suitable crop conditions, and safe operation help improve output quality while reducing blockages and injury risks.

Main parts

- Feed chute guides crop residue into the machine.
- Cutting or chopping mechanism processes the straw.
- Rotor or drum carries the cutting components.
- Concave or sieve separates processed material according to the machine design.
- Blower or conveying system moves processed straw toward the outlet.
- Collection chute directs straw into a trailer or collection area.
- Drive system receives power from the tractor or machine engine.
- Guards and adjustment controls protect and regulate operating components.

Before you start

1. Read the manual and confirm the straw reaper is suitable for the crop and operating conditions.
2. Inspect the frame, hitch, guards, belts, blades, fasteners, and drive components for damage.
3. Check blade condition and ensure all guards are correctly fitted.
4. Verify compatibility with the tractor or power source and confirm safe connections.
5. Clear the field of stones, wire, metal, and other foreign objects where practical.
6. Keep bystanders away and wear suitable PPE, including eye and hearing protection.

How to use

1. Park the tractor or machine safely and isolate the power before inspection or adjustment.
2. Connect the machine according to the manufacturer's instructions and secure all required pins and guards.
3. Set the cutting height and other adjustments for the crop and field conditions.
4. Start the power source and engage the drive using the specified sequence.
5. Allow the machine to reach the recommended operating speed before feeding crop residue.
6. Move forward at a controlled speed and monitor the feed, straw quality, and machine load.
7. Stop feeding and disengage the drive if blockage, excessive vibration, or unusual noise occurs.
8. Wait for all moving parts to stop and isolate power before clearing a blockage.
9. Collect the processed straw safely and avoid approaching the discharge outlet while material is being ejected.

10. At the end, disengage the drive, shut down the machine, clean residues, and inspect for damage.

⚠ Safety

- Never place hands or feet into the feed chute or discharge outlet.
- Keep guards fitted and never approach rotating shafts, belts, blades, or drums.
- Switch off the engine, isolate power, and wait for all parts to stop before clearing blockages.
- Do not wear loose clothing or operate the machine while tired or distracted.
- Keep people away from the discharge area and avoid directing material toward bystanders.
- Do not operate with damaged blades, loose fasteners, missing guards, or an unstable tractor connection.

Care & storage

- Remove straw, dust, and crop residue after use using the approved method.
- Inspect blades, belts, bearings, guards, and fasteners for wear or damage.
- Lubricate specified points and maintain the drive system according to the manual.
- Store the machine in a dry, secure location and record faults requiring repair.

Common mistakes to avoid

- Operating with damaged blades or missing guards.
- Feeding crop residue too quickly and overloading the machine.
- Attempting to clear a blockage before all moving parts stop.
- Allowing bystanders near the feed or discharge area.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.